Business Data Mining Homework Session 5

16.2 Performance on Training and Validation Data. Two different models were fit to the same time series. The first 100 time periods were used for the training set and the last 12 periods were treated as a hold-out set. Assume that both models make sense practically and fit the data pretty well. Below are the RMSE values for each of the models:

a. Which model appears more useful for explaining the different components of this time series? Why?

I think Model B is more useful for explaining the different components, and here's why.

Model A performs better than B on the training set, but worse on the validation set. I think this means Model A is probably overfitting. It learned too much of the noise, which doesn't really help in explaining the important components like the actual trend and seasonality.

On the other hand, the difference in RMSE between the training and validation sets for Model B is much smaller (675 - 669 = 6). Plus, Model B's validation set RMSE is actually lower than Model A's. This shows that Model B isn't overfitting—it's more stable and generalizable.

Also, since RMSE tends to penalize larger errors more heavily, paying attention to RMSE is really important here.

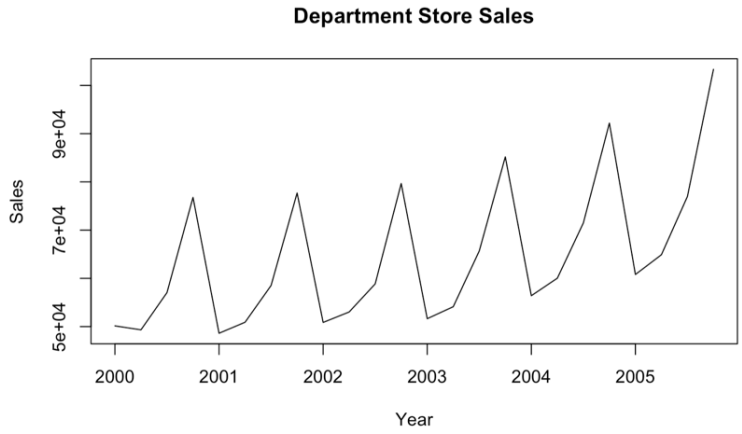
b. Which model appears to be more useful for forecasting purposes? Why?

I think Model B is more useful for forecasting. Here's why.

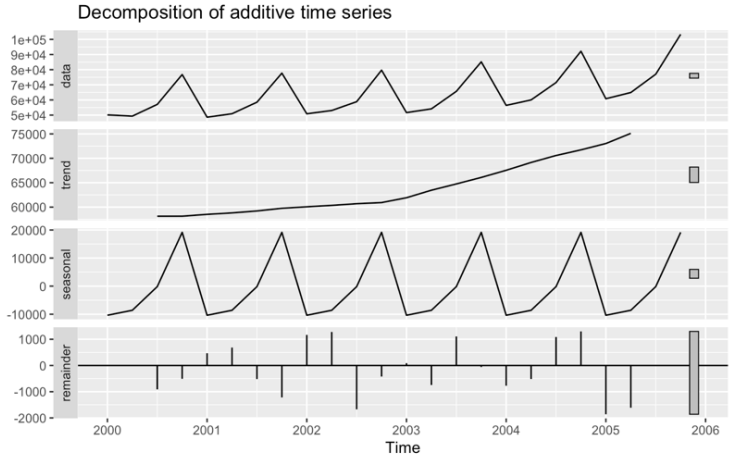
Based on our previous discussion, Model B isn't overfitting and captures the real trend and seasonality better. The RMSE difference between its training and validation sets is really small, and its validation RMSE is actually lower than Model A's. This all suggests Model B will perform better when we use it to forecast future values.

16.3 Forecasting Department Store Sales. The file DepartmentStoreSales.csv contains data on the quarterly sales for a department store over a 6-year period (data courtesy of Chris Albright).

a. Create a well-formatted time plot of the data.



b. Which of the four components (level, trend, seasonality, noise) seem to be present in this series?



According to the plot, all 4 components are present in the series.

Level: This is represented by the overall height of the series, which is captured in the Trend component.

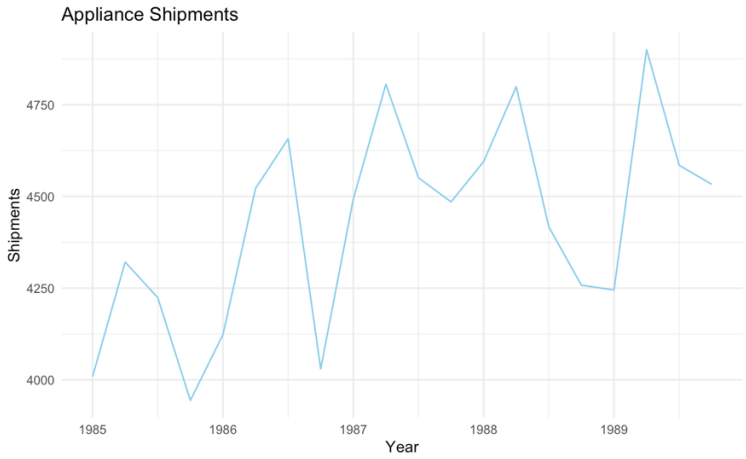
Trend: I think it is a multiplicative trend. We can see a clear upward trend over the years.

Seasonality: I think it is additive seasonality. Sales consistently peak around the end of each year (likely holiday season) and drop to a low point in the first quarter.

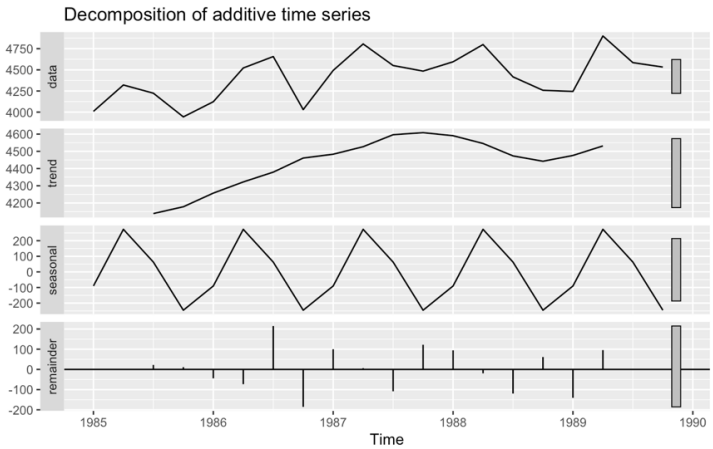
Noise: We can see small, irregular fluctuations in the Random component. However, the magnitude is very small compared to the trend and seasonality, indicating relatively low noise.

16.4 Shipments of Household Appliances. The file ApplianceShipments.csv contains the series of quarterly shipments (in million $) of US household appliances between 1985 and 1989 (data courtesy of Ken Black).

a. Create a well-formatted time plot of the data.



b. Which of the four components (level, trend, seasonality, noise) seem to be present in this series?



According to the plot, all four components are present in the series.

Level: The level is clearly visible as the overall baseline around which the series fluctuates. It's best seen in the upward shift of the entire data between the first and second halves of the "Observed" series.

Trend: The trend component shows a clear pattern of increase, followed by a slight decline, and then another increase. This non-linear movement is distinct and forms the main long-term direction of the data.

Seasonality: There is a consistent additive seasonality pattern visible. The seasonal component shows regular peaks during summer months and troughs during winter, repeating stably each year with roughly constant amplitude.

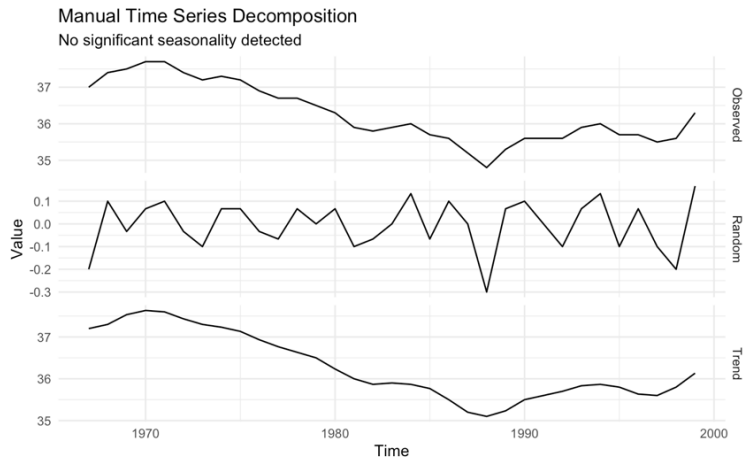
Noise: The random component demonstrates that noise has a more significant impact on the seasonal pattern than on the trend. While the trend remains relatively smooth, the seasonal values show noticeable irregular fluctuations around their typical seasonal path.

16.5 Canadian Manufacturing Workers Workhours. The time plot in Figure 16.5 describes the average annual number of weekly hours spent by Canadian manufacturing workers (data are available in CanadianWorkHours.csv-----thanks to Ken Black for the data).

a. Reproduce the time plot.

b. Which of the four components (level, trend, seasonality, noise) seem to be present in this series?

I checked the dataset, and found the frequency is 1, which means we cannot use stl() or decompose() in R language, and we have the reason to think that the data has no seasonality. Therefore, I manually decomposed it into trend and noise components, visualizing them with ggplot2().



The chart shows that level, trend and noise are present in the series.

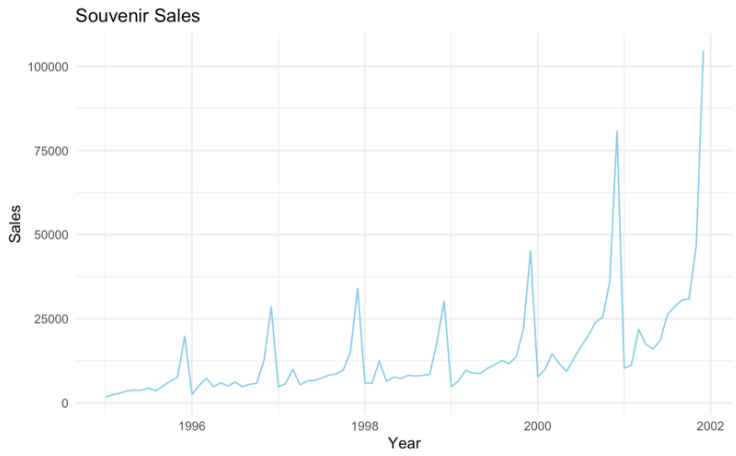
Level: The level is present and is represented by the overall baseline of the data, approximately between 35 and 37 on the y-axis.

Trend: A non-monotonic trend is clearly visible. It shows a general downward movement for the first half of the series, followed by a slight upward movement in the latter part. This makes it difficult to fit a simple linear trend.

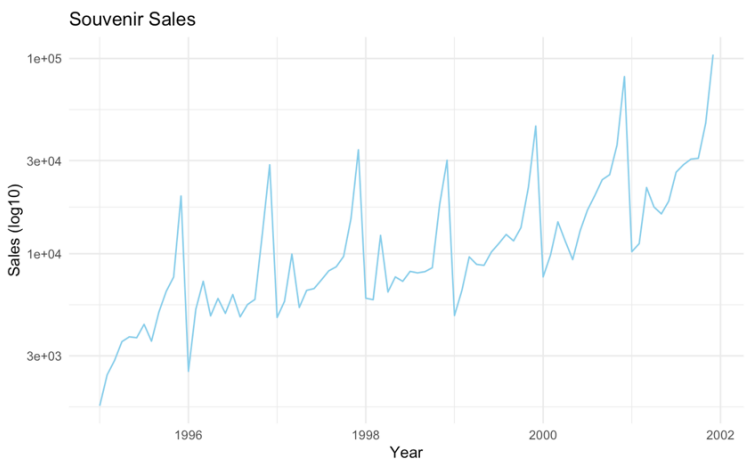
Noise: Noise is present, as seen in the Random component. However, its magnitude is relatively small (values range roughly from -0.3 to +0.1) compared to the scale of the original data, indicating it has a minor influence on the series.

16.6 Souvenir Sales: The file SouvenirSales.csv contains monthly sales for a souvenir shop at a beach resort town in Queensland, Australia, between 1995-2001.
Back in 2001, the store wanted to use the data to forecast sales for the next 12 months (year 2002). They hired an analyst to generate forecasts. The analyst first partitioned the data into training and validation sets, with the validation set containing the last 12 months of data (year 2001). She then fit a regression model to sales, using the training set.

a. Create a well-formatted time plot of the data.



b. Change the scale on the x-axis, or on the y-axis, or on both to log-scale in order to achieve a linear relationship. Select the time plot that seems most linear.



To achieve the most linear relationship, I experimented with both y-axis and double logarithmic transformations. I found that applying a logarithmic transformation only to the Y-axis resulted in the most linear appearance of the time series.

The plot with the log10 Y-scale transforms the previously exponential looking growth pattern into a nearly straight line. This indicates a stable relative growth rate over time. In contrast, the double logarithmic plot did not improve the linearity and unnecessarily distorted the time axis.

c. Comparing the two time plots, what can be said about the type of trend in the data?

Both plots clearly show a strong upward trend and regular seasonality. The key difference lies in how these components behave.

Before Log Transformation (Original Scale): The pattern exhibits a multiplicative structure. This is evident because the seasonal peaks and troughs (the "waves") increase in magnitude as the overall trend rises. The fluctuations are proportional to the level of the data.

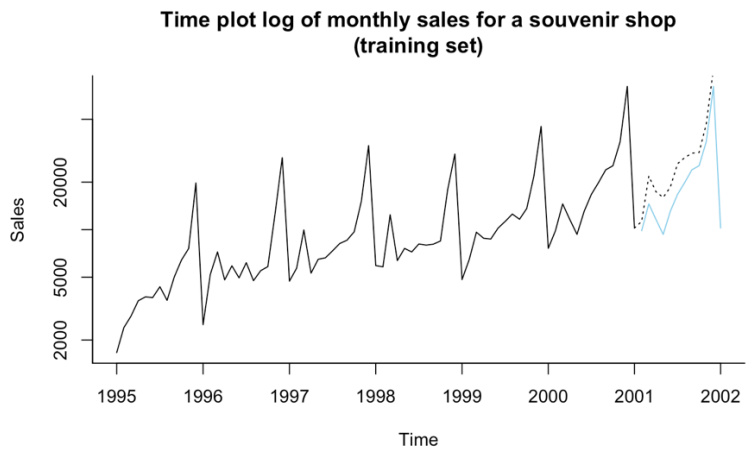
After Log Transformation (Log Scale): The series becomes additive. The seasonal fluctuations now have a constant magnitude throughout the entire period, independent of the trend's level. This transformation successfully stabilizes the variance, making the pattern easier to model.

In both views, the seasonal peak consistently occurs around the winter, likely driven by Christmas promotions, confirming the timing of the main seasonal effect remains unchanged.

d. Why were the data partitioned? Partition the data into the training and validation set as explained above.

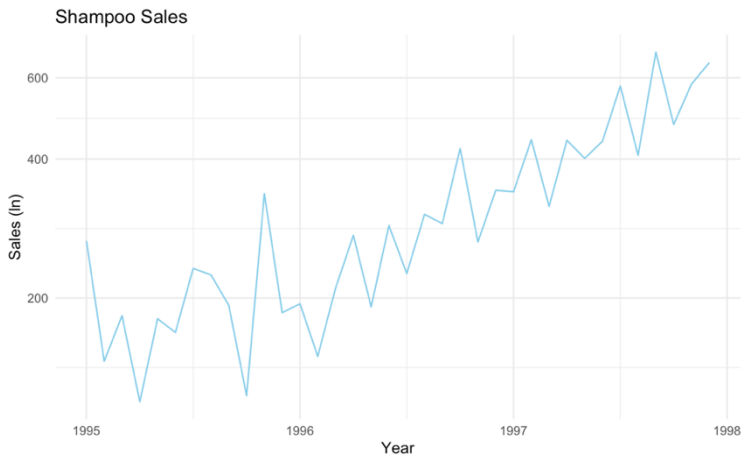
In forecasting problems, data was partitioned into training and validation sets to evaluate the model's performance on unseen data.

Training period: 1995 year 1 to 2000 year 12
Validation period: 2001 year 1 to 2001 year 12
Total observations: 84 = Training: 72 + Validation: 12

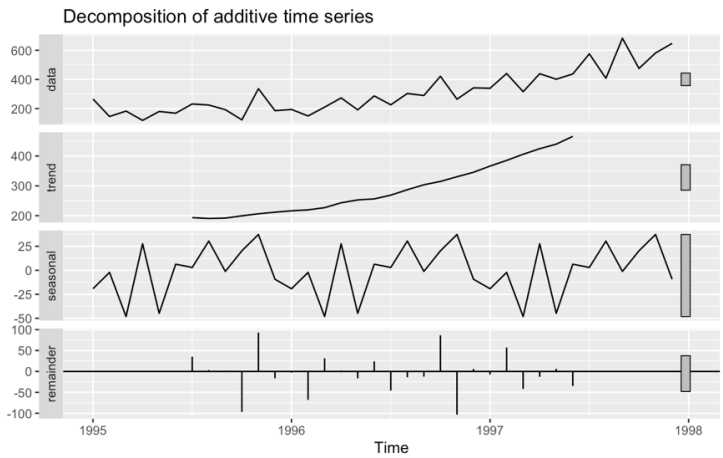


16.7 Shampoo Sales. The file ShampooSales.csv contains data on the monthly sales of a certain shampoo over a 3-year period.

a. Create a well-formatted time plot of the data.



b. Which of the four components (level, trend, seasonality, noise) seem to be present in this series?



Based on the plot, all 4 components are present.

Level: The level is represented by the overall baseline of the data, visible in the trend component, which fluctuates around a value of approximately 200.

Trend: I think it is a multiplicative trend. A clear upward trend is present, showing that sales have generally increased over time.

Seasonality: There is an additive seasonality. Although the seasonal component appears chaotic at first glance, its fluctuation pattern is highly repetitive each year.

Noise: The random component shows noticeable fluctuations, indicating that noise significantly affects the seasonal pattern. However, the noise magnitude is relatively small compared to the trend and seasonality.

c. Do you expect to see seasonality in sales of shampoo? Why?

Yes, I expect clear seasonality. Shampoo is a staple product, but its sales are driven by seasonal factors like. Holiday peaks (Christmas for gift sets and promotions). Summer increases due to more frequent washing from heat and vacation travel. These predictable patterns create regular fluctuations in sales throughout the year.

d. If the goal is forecasting sales in future months, which of the following steps should be taken?

①Partition the data into training and validation sets?

Yes. This is an essential first step to a model's predictive performance on unseen data.

② Tweak the model parameters to obtain good fit to the validation data?

No. Doing so would lead to data leakage and overfitting to the validation set, undermining its role as an independent benchmark for generalization.

③ Look at MAPE and RMSE values for the training set?

No, although a model fit the training set very well, it may have terrible performance at validation data.

④ Look at MAPE and RMSE values for the validation set?

Yes, according to the error, we could find out how good the model could forecast the future. These metrics provide the best available estimate of the model's future forecasting accuracy, as they reflect performance on data not used during model training.

11.3 Car Sales. Consider the data on used cars (ToyotaCorolla.csv) with 1436 records and details on 38 attributes, including Price, Age, KM, HP, and other specifications. The goal is to predict the price of a used Toyota Corolla based on its specifications.

Neural Network RMSE Results (set seed is 123)		
Model	RMSE_Train	RMSE_Test
1 layer, 2 nodes	0.0381747	0.0385008
1 layer, 5 nodes	0.0341291	0.0404328
2 layers, 5 nodes each	0.0347475	0.0377051

i. What happens to the RMS error for the training data as the number of layers and nodes increases?

As the number of layers and nodes increases, we could find that the RMSE for the training data decrease. It demonstrates the neural network's enhanced learning capacity to capture complex patterns. However, this improvement primarily reflects better performance on seen data and does not guarantee superior generalization to new data. If the RMSE on the validation set stops decreasing or starts to rise while the training error falls, it is a clear indicator of overfitting, meaning the model is memorizing noise rather than learning the underlying relationship.

ii. What happens to the RMS error for the validation data?

The RMSE for the validation set does not show a consistent pattern of increase. Instead, it fluctuates. It increases from the 2-node to the 5-node model but then decreases again for the more complex 2-layer model, which achieved the lowest test error.
This indicates that while overfitting is a risk, increasing model complexity can sometimes lead to better generalization if the additional capacity is used to learn the true underlying pattern rather than just memorizing noise. The best model for forecasting is the one with the lowest validation error.

iii. Comment on the appropriate number of layers and nodes for this application.

I think 2 layers and 5 nodes is the most appropriate compare to others. Because it has the lowest RMSE of validation data. Although it has the risk of overfitting, the additional layer in the 2-layer architecture appears to provide better generalization capability. This suggests that the increased complexity helps capture the underlying patterns in the data without merely memorizing noise. 1 layer and 2 nodes looks not bad, but it may be too simple to capture full information.